

CRUD Operations

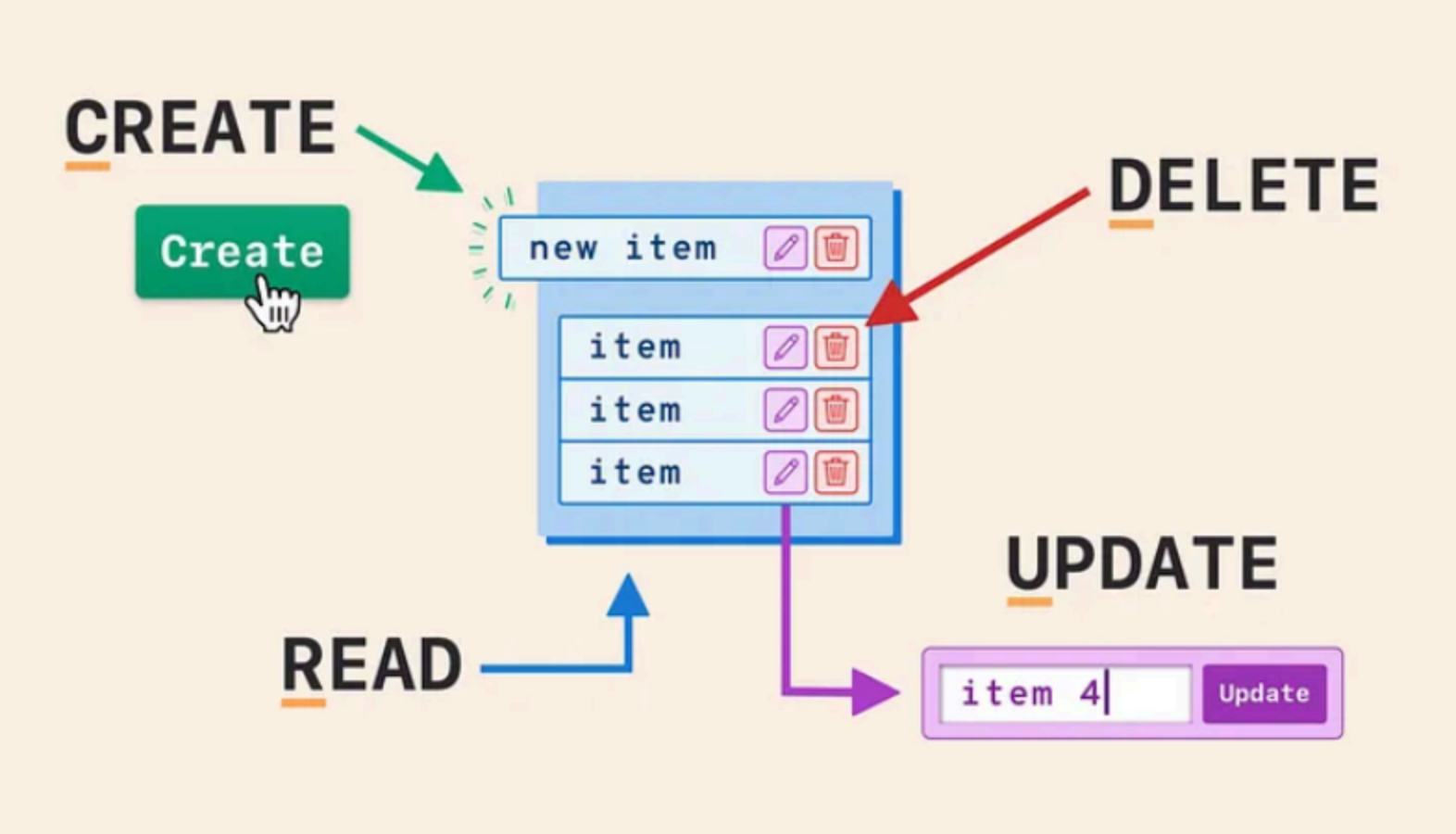
SWEN1 | BIF3

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What is CRUD?


CREATE
Create
new item
item
item
item
DELETE
UPDATE
item 4
Update
READ

CRUD = Create, Read, Update, Delete

Core operations of any database-driven application:

Operation	SQL Keyword	Description
Create	`INSERT`	Add new rows to a table
Read	`SELECT`	Retrieve data from a table
Update	`UPDATE`	Modify existing data
Delete	`DELETE`	Remove data from a table

You will now implement all four using code in Java and C#.

Required Setup

✅ PostgreSQL container running
✅ `app_db` or `postgres` as your database
✅ Existing table: `users(id SERIAL PRIMARY KEY, name TEXT)`

> 📌 You may create this table manually using `psql`:

```
CREATE TABLE users (
  id SERIAL PRIMARY KEY,
  name TEXT
);
```



SQL Commands for CRUD

CREATE (Insert)

```sql
INSERT INTO users (name) VALUES ('Alice');
```

READ (Select)

```sql
SELECT * FROM users;
```

UPDATE

```sql
UPDATE users SET name = 'Bob' WHERE id = 1;
```

DELETE

```sql
DELETE FROM users WHERE id = 1;
```



Java: JDBC Implementation

INSERT (Create)

```
String sql = "INSERT INTO users (name) VALUES (?)";

try (PreparedStatement stmt = conn.prepareStatement(sql)) {
    stmt.setString(1, "Alice");
    stmt.executeUpdate();
}
```

SELECT (Read)

```
String sql = "SELECT * FROM users";

try (Statement stmt = conn.createStatement();
     ResultSet rs = stmt.executeQuery(sql)) {

    while (rs.next()) {
        System.out.println(rs.getInt("id") + ": " + rs.getString("name"));
    }
}
```

UPDATE

```
String sql = "UPDATE users SET name = ? WHERE id = ?";

try (PreparedStatement stmt = conn.prepareStatement(sql)) {
    stmt.setString(1, "Bob");
    stmt.setInt(2, 1);
    stmt.executeUpdate();
}
```

DELETE

```
String sql = "DELETE FROM users WHERE id = ?";

try (PreparedStatement stmt = conn.prepareStatement(sql)) {
    stmt.setInt(1, 1);
    stmt.executeUpdate();
}
```



C#: ADO.NET Implementation

INSERT (Create)

```
var sql = "INSERT INTO users (name) VALUES (@name)";
using var cmd = new NpgsqlCommand(sql, conn);
cmd.Parameters.AddWithValue("name", "Alice");
cmd.ExecuteNonQuery();
```

SELECT (Read)

```
var sql = "SELECT * FROM users";
using var cmd = new NpgsqlCommand(sql, conn);
using var reader = cmd.ExecuteReader();

while (reader.Read()) {
    Console.WriteLine($"{reader.GetInt32(0)}: {reader.GetString(1)}");
}
```

UPDATE

```
var sql = "UPDATE users SET name = @name WHERE id = @id";
using var cmd = new NpgsqlCommand(sql, conn);
cmd.Parameters.AddWithValue("name", "Bob");
cmd.Parameters.AddWithValue("id", 1);
cmd.ExecuteNonQuery();
```

DELETE

```
var sql = "DELETE FROM users WHERE id = @id";
using var cmd = new NpgsqlCommand(sql, conn);
cmd.Parameters.AddWithValue("id", 1);
cmd.ExecuteNonQuery();
```



Tips, Errors & Summary

Common Mistakes

❌ Table does not exist → Create it first
❌ NullReferenceException or SQLException → Connection not opened
❌ SQL syntax error → Copy/paste directly into psql to test

Debugging Advice

Java:

```
e.printStackTrace();
```

C#:

```
Console.WriteLine(ex.Message);
```

Good Practice

✅ Use PreparedStatement / NpgsqlCommand with parameters
✅ Always try-with-resources / using
✅ Never concatenate user input into SQL strings

Summary

- You now implemented all CRUD operations from code
- Both Java and C# use SQL statements via libraries
- Understanding SQL helps you debug and extend your app

Further Reading

- JDBC SQL Execution (Oracle Tutorial)
- Npgsql Command API (ADO.NET for PostgreSQL)
- PostgreSQL SQL Command Reference